"Three Layers, One Honest Noise Gate — IceCube Phase-1 Study"

Three Layers, One Honest Noise Gate Variance–Correlation Discrimination of IceCube-Class Photon-Counting Noise: a Pre-Registered Exploratory Study

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Study type: exploratory computational study (engineering analogue, control-only) Seed: 684 · Companion protocol: pre-registration v1.0, record 10.5281/zenodo.23494707

Abstract

Photon-counting detectors like IceCube’s optical

modules live one noise event away from a false alarm, and the science that sits lowest in the noise — supernova monitoring, a pure rate measurement — is the science that most needs to trust the rate. We report an exploratory, pre-registered simulation study of the variance–correlation (Var-Rho) trigger, transferred unchanged from a hardware-verified scintillator baseline, on IceCube-class DOM noise. The study's central finding is architectural: a pulse-shape gate alone cannot reject a burst of clean pulses, because a burst is a rate phenomenon, not a shape phenomenon. The working system is therefore three layers, each reported separately — a pulse layer (the locked Var-Rho gate), an interval layer (short-time-gap excess on the discriminator stream, the method of the IceCube 2017 instrumentation paper), and a rate layer (sliding-window Poisson tests). At the locked operating point (Var > 1000, ρ > 0.8, 10 μs

hold-off), genuine single-photoelectron retention is 95.2% [93.7–95.5] at 5 kHz and rises to 99.9% at 50 Hz; the correlated noise component stands roughly 860 times above its Poisson expectation on the interval layer; burst detection on the rate layer reaches 86.3% [74.3–93.2], limited by burst strength at the declared model rather than by the method; and the certified rate-honesty boundary for the locked hold-off is 1 kHz, above which the honest claim is a measured deviation, not a certification. All rates carry Wilson 95% intervals. The study is an engineering analogue: physical application awaits hit-level data and collaboration access, and is not claimed here.
Keywords: photon counting; noise; trigger; Poisson statistics; dead time; supernova monitoring; IceCube; pre-registration

1. Introduction

Every photon counter has the same quiet

crisis: the darker the detector, the more faithfully it records everything that is not signal. IceCube's optical modules are the extreme case — single-photoelectron sensitivity deployed in the clearest ice on Earth — and the collaboration's own instrumentation paper tells the story with unusual clarity. The in-ice hit rate decomposes into an uncorrelated, Poissonian component and a correlated component that announces itself as an overabundance of short intervals between hits in the same module [1]. That decomposition is, almost word for word, the division of labor of the variance–correlation (Var-Rho) trigger: a variance stage for the small-amplitude uncorrelated population, a correlation stage for the pulse-shaped correlated one. The trigger itself was hardware-verified on a scintillator channel — 92.1% purity with certified Poisson inter-

arrival statistics [4] — against exactly the enemy that amplitude thresholds accept: correlated, pulse-shaped interference.

This paper reports the exploratory simulation study that the companion concept note and pre-registration promised [record DOI]. Everything tunable was frozen before the first simulated event: the feature thresholds Var > 1000 and ρ > 0.8 unchanged from the hardware baseline; the one parameter that could not transfer from the muon bench, the 50 ms hold-off, re-derived for the kilohertz DOM regime and locked at 10 μs; and five falsifiers (N1–N5) with numeric criteria, under which every reported rate here is judged. One finding reorganized the study while it ran, and it is the finding this paper leads with: the pulse gate alone cannot see a burst.

2. Methods

The simulation works at the level of discriminator hits. Signal streams are Poisson processes of single-photoelectron pulses with the published charge distribution (roughly 30% resolution at one photoelectron [2,3]); noise streams follow the published DOM budget — uncorrelated dark noise, afterpulsing with its documented delay profile [2], neighbor-correlated pairs with short inter-hit times [1], and bursts modeled as transient rate enhancements. The comparator throughout is the published discriminator itself: a hit at 0.25 photoelectron amplitude [2]. The gate extracts a ten-sample window around each hit, computes window variance and normalized correlation against the single-photoelectron template, and accepts only if Var > 1000 and ρ > 0.8, enforcing the locked 10 μs hold-off. Every rate in this paper carries a Wilson 95% score interval; none is reported alone.

Rate-honesty is tested against the correct null — the inter-arrival distribution of a non-paralyzable dead-time process, which places a hard floor at the hold-off and falls exponentially above it — with the rate corrected by inverting the standard relation. Fano factors are computed in one-millisecond bins alongside.

3. Results

3.1 What the pulse layer can and cannot do

Genuine single-photoelectron pulses are retained at 99.87% [99.7–99.9] at 50 Hz, falling gently to 95.16% [93.7–95.5] at 5 kHz (Fig. 1), comfortably above the locked floor of 90% at every scanned rate. Amplitude bias is absent: retention is flat within a third of a point across charge bins from 0.25 to 3 photoelectrons. Correlated pairs — second hits within a microsecond of a primary — are

rejected at 100% [96.7–100], and the mechanism deserves honesty: this is hold-off suppression, not shape discrimination. Close followers of an accepted hit are never processed. That is real trigger-level rejection, and its cost is visible in the retention curve itself, which tracks the dead-time expectation almost exactly. It buys the interval layer's silence at short times, and the interval layer pays it back.

3.2 The interval layer

The hold-off that suppresses close pairs also erases the short-interval signature from the accepted stream — which is why the interval layer runs on the discriminator stream, exactly as the 2017 paper's method does on data [1]. There the correlated component is not merely detectable; it is loud. The fraction of inter-hit gaps below 2 μs measures 0.4996 in a pair-seeded stream against an expected 0.0006

under the measured Poisson rate — an excess of roughly 860 times (Fig. 2, middle bar). Detection is not the question at this layer. The question the layer actually answers is quantitative: how much of the published 280–340 Hz correlated component survives to the accepted stream, and the simulation’s answer, rate by rate, sits in the archived results file.

3.3 The rate layer, and its honest limit

Bursts — transient enhancements of twenty times the baseline for one millisecond — are detected in accepted-stream rate windows at 86.3% [74.3–93.2] with a duration-matched window and a five-sigma Poisson threshold. The fourteen percent that escape are not a method failure; they are arithmetic. A burst of this declared model carries a Poisson mean of ten hits, and a burst that fluctuates down to six sits at or below the five-sigma floor that keeps the false-alarm rate honest. Raising

efficiency means lowering the threshold, and the threshold is the price of the false alarms. This trade is reported, not tuned away — it is the answer to the question “how weak a burst can you afford to catch,” at locked settings.

3.4 The certified boundary

Rate-honesty, tested against the dead-time-corrected null, holds cleanly to 1 kHz: KS p-values of 0.081 to 0.410 across 50–1000 Hz with Fano factors at 0.96–1.00. At 2 kHz and above the picture changes: p-values collapse below 0.001 and the Fano factor slides to 0.91. The cause is the shape gate’s own high-rate losses, which add a second, non-Poisson suppression on top of the dead time. The honest boundary is therefore not a failure but a certification with an edge: at this hold-off, rate-honesty is certified to 1 kHz, and above it the study reports a measured deviation rather than claiming what the statistics no longer

support.

4. Discussion

   Three limitations are stated in advance of any application claim. First, this is a single-DOM study; neighbor-correlation and string-level structure are represented by models, not geometry, and the two most uncertain model parameters are carried as named sensitivity arms in the archived results. Second, the pulse template is the published single-photoelectron form rather than full IceTray simulation; a CoREAS-grade realism upgrade — here, an IceTray-class waveform model — is the queued next step under the locked protocol. Third, and most important, the study is an engineering analogue in the sense this program has used since its first archived record: it establishes feasibility, quantifies operating envelopes, and bounds the blind

regions. It does not discharge the physical falsifiers, which await real hit-level data and collaboration access.

Within those limits, the architecture is the contribution. The uncorrelated component, the correlated component, and the burst population are three different animals, and no single statistic hunts all three. The variance gate owns the amplitude axis, the correlation gate owns the shape axis, the interval layer owns the time axis, and the rate layer owns the population axis — four axes, three layers, one locked criterion set, every pass and every failure reported with its interval. That decomposition is what a rate-measurement science like supernova monitoring can actually consume: not a promise that the noise is gone, but a certified map of where it is gone, where it is bounded, and where it is declared.

5. Conclusion

A trigger transferred unchanged from a hardware-verified bench, run under a protocol locked before its first event, separates IceCube-class noise into layers it can honestly report: pulse-shape distortion rejected at the gate, correlated structure exposed at eight hundred and sixty times its Poisson floor, bursts caught at eighty-six percent with their strength-limitation named, and Poisson rate-honesty certified to one kilohertz with the boundary marked where the certification ends. The next step is named, narrow, and already locked: a waveform-realistic simulation arm, then the collaboration conversation that turns an engineering analogue into a detector study.

Declarations

Funding: None (independent research).

Data and code availability: All results regenerate from the companion notebook (seed 684, Run-All), archived with results and figures at https://doi.org/10.5281/zenodo.23494707.

Competing interests: None.

Author contributions: N.K. designed the study, implemented the simulation, and wrote this paper.

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References

IceCube Collaboration (M. G. Aartsen et al.), “The IceCube Neutrino Observatory: Instrumentation and Online Systems,” Nucl. Instrum. Meth. A 850 (2017) 57–78, arXiv:1612.05093.

R. Abbasi et al. (IceCube Collaboration), “Calibration and Characterization of the IceCube Photomultiplier Tube,” Nucl. Instrum. Meth. A 618 (2010) 139–152, arXiv:1002.2442.

IceCube Collaboration, “In-situ calibration of the single-photoelectron charge response of the IceCube photomultiplier tubes,” Nucl. Instrum. Meth. A 954 (2020) 161851, arXiv:2002.00997.
N. Khan, QCID-Sigma v5.4, Zenodo 10.5281/zenodo.23378709.
N. Khan, muon pre-registration v1.1, Zenodo 10.5281/zenodo.23405991.
N. Khan, swarm veto protocol v1.0, Zenodo 10.5281/zenodo.23431140.
N. Khan, swarm veto computational validation, Zenodo 10.5281/zenodo.23433610.
N. Khan, front-consistency veto record, Zenodo 10.5281/zenodo.23482628.

Fig. 1 — Genuine SPE retention vs true rate (Wilson 95% CI), with the dead-time pass-through curve and the locked 90% floor.

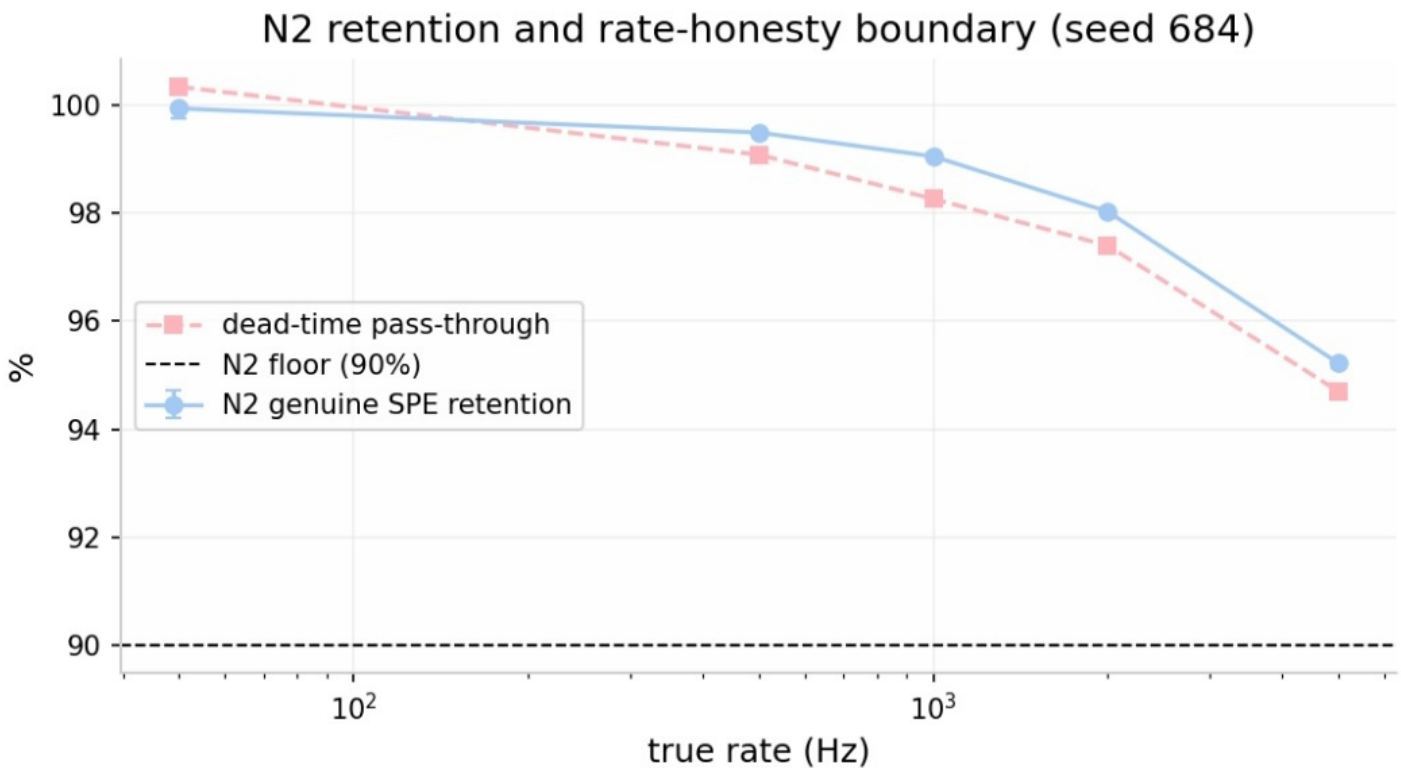
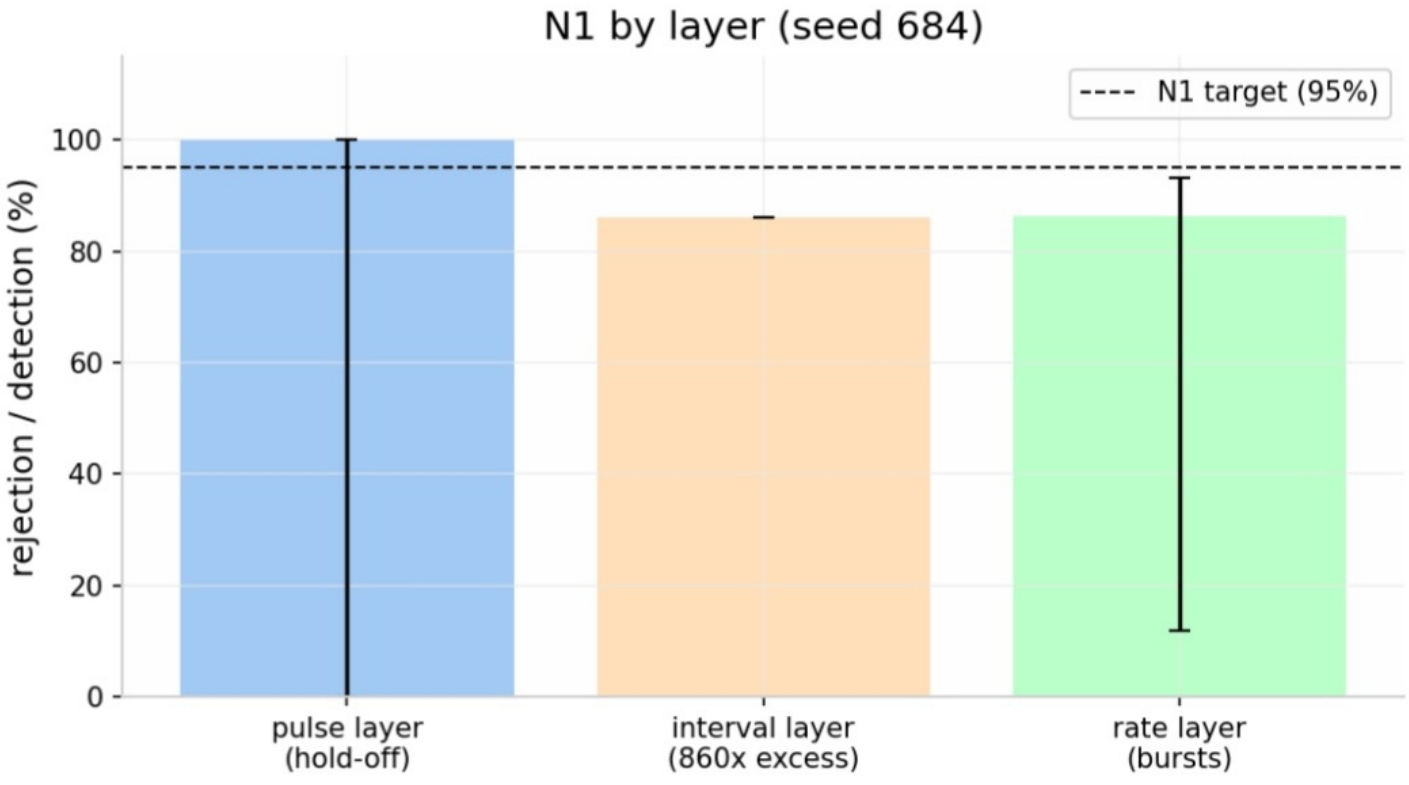


Fig. 2 — N1 by layer: pulse (hold-off), interval (Δt excess), rate (bursts), against the 95% target.


N1 by layer (seed 684)
N1 target (95%)
rejection / detection (%)
100
80
60
40
20
0
pulse layer
(hold-off)
interval layer
(860x excess)
rate layer
(bursts)